

MATHEMATICS 2270
Quiz 3, Fall semester 2012

$$A = \begin{bmatrix} 1 & 1 & 2 & 0 & 1 \\ 2 & 0 & 3 & 1 & 0 \\ 1 & 3 & 3 & -1 & 3 \end{bmatrix}$$

Name:

1. Find the null space of the matrix $A = (1, 1, 2, 0, 1 | 2, 0, 3, 1, 0 | 1, 3, 3, -1, 3)$.

$$\begin{array}{ccccc} 1 & 1 & 2 & 0 & 1 \\ 2 & 0 & 3 & 1 & 0 \\ 1 & 3 & 3 & -1 & 3 \end{array} \rightarrow \begin{array}{ccccc} 1 & 1 & 2 & 0 & 1 \\ 0 & -2 & -1 & 1 & -2 \\ 0 & 2 & 1 & -1 & 2 \end{array} \rightarrow \begin{array}{ccccc} 1 & 1 & 2 & 0 & 1 \\ 0 & 1 & 1/2 & -1/2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \rightarrow \begin{array}{ccccc} 1 & 0 & 3/2 & 1/2 & 0 \\ 0 & 1 & 1/2 & -1/2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}$$

$$x = -3/2 z - 1/2 u$$

$$y = -1/2 z + 1/2 u - v$$

$$\begin{pmatrix} -3/2 z - 1/2 u \\ -1/2 z + 1/2 u - v \\ z \\ u \\ v \end{pmatrix} = z \begin{pmatrix} -3/2 \\ -1/2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + u \begin{pmatrix} -1/2 \\ 1/2 \\ 0 \\ 1 \\ 0 \end{pmatrix} + v \begin{pmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

2. Let U be a subset of a vector space V . When is U a sub-vector space of V ?

① $0 \in U$

② If $u_1, u_2 \in U$ then $u_1 + u_2 \in U$

③ If $u \in U$ & $\lambda \in \mathbb{R}$ then $\lambda u \in U$

3. Let U_1, U_2 be subspaces of a vector space V . Show that $W = U_1 + U_2 := \{u_1 + u_2 \mid u_1 \in U_1, \text{ and } u_2 \in U_2\}$ is a subspace of V .

① $\begin{cases} 0 \in U_1 \\ 0 \in U_2 \end{cases}$ because U_1 & U_2 are subspaces.

but then $0 = 0 + 0 \in U_1 + U_2 = W$

② If $\cancel{w_1}, \cancel{w_2} \in \cancel{W}$ then $\begin{cases} \cancel{w_1} = \cancel{u_1} + \cancel{u_2} \\ \cancel{w_2} = \cancel{u_1} + \cancel{u_2} \end{cases}$ where $\cancel{w_1}, \cancel{u_1} \in U_1$
 $\cancel{w_2}, \cancel{u_2} \in U_2$

So $\cancel{w_1} + \cancel{u_1} \in U_1$ & $\cancel{w_2} + \cancel{u_2} \in U_2$

but then $\cancel{w_1} + \cancel{w_2} = \cancel{u_1} + \cancel{u_2} + \cancel{u_1} + \cancel{u_2} = (\cancel{u_1} + \cancel{u_1}) + (\cancel{u_2} + \cancel{u_2}) \in U_1 + U_2$
 $= W$

③ If $w \in W$ & $\lambda \in \mathbb{R}$, then $w = w_1 + w_2$ where $w_1 \in U_1$
 $w_2 \in U_2$

So $\begin{matrix} \lambda w_1 \in U_1 \\ \lambda w_2 \in U_2 \end{matrix}$ because U_1 & U_2 are subspaces

So $\lambda w = \lambda(w_1 + w_2) = \lambda w_1 + \lambda w_2 \in U_1 + U_2 = W$